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 NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Data Science for Engineers (course)


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 Course
outline

 About
NPTEL ()

 How does an
NPTEL
online
course
work? ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

☐ Multivariate
Optimization
With Equality
Constraints
(unit?)

Week 5 : Assignment 5

The due date for submitting this assignment has passed.

Due on 2025-08-27, 23:59 IST.

Assignment submitted on 2025-08-27, 21:46 IST

 1) The values of μ_1 , μ_2 and μ_3 while evaluating the Karush-Kuhn-Tucker (KKT) condition with all the constraints being inactive are **1 point**

- ☐ $\mu_1 = \mu_2 = \mu_3 = 1$
☒ $\mu_1 = \mu_2 = \mu_3 = 0$
☐ $\mu_1 = \mu_3 = 0, \mu_2 = 1$
☐ $\mu_1 = \mu_2 = 0, \mu_3 = 1$

Yes, the answer is correct.

Score: 1

Accepted Answers:

 $\mu_1 = \mu_2 = \mu_3 = 0$

 2) Gradient based algorithm methods compute **1 point**

- ☐ only step length at each iteration
☒ both direction and step length at each iteration
☐ only direction at each iteration
☐ none of the above

Yes, the answer is correct.

Score: 1

Accepted Answers:

both direction and step length at each iteration

unit=63&less
n=64)

☐ Multivariate
Optimization
With Inequality
Constraints
(unit?

unit=63&less
n=65)

☐ Introduction to
Data Science
(unit?
unit=63&less
n=66)

☐ Solving Data
Analysis
Problems - A
Guided
Thought
Process (unit?
unit=63&less
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☐ Dataset (unit?
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☐ FAQ (unit?
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☐ Week 5
Feedback
Form : Data
Science for
Engineers!!
(unit?
unit=63&less
n=157)

☒ **Quiz: Week 5
: Assignment
5
(assessment?
name=240)**

Week 6 ()

Week 7 ()

**Text
Transcripts
()**

3) The point on the plane $x + y - 2z = 6$ that is closest to the origin is

1 point

- ☐ (0, 0, 0)
☐ (1, 1, 1)
☐ (-1, 1, 2)
☒ (1, 1, -2)

Yes, the answer is correct.

Score: 1

Accepted Answers:

(1, 1, -2)

4) Find the maximum value of $f(x, y) = 49 - x^2 - y^2$ subject to the constraints $x + 3y = 10$.

1 point

- ☐ 49
☐ 46
☐ 59
☒ 39

Yes, the answer is correct.

Score: 1

Accepted Answers:

39

5) The minimum value of $f(x, y) = x^2 + 4y^2 - 2x + 8y$ subject to the constraint $x + 2y = 7$ occurs at the below point:

1 point

- ☐ (5, 5)
☐ (-5, 5)
☐ (1, 5)
☒ (5, 1)

Yes, the answer is correct.

Score: 1

Accepted Answers:

(5, 1)

6) Which of the following statements is/are **NOT TRUE** with respect to the multi variate optimization? **1 point**

- I - The gradient of a function at a point is parallel to the contours
II - Gradient points in the direction of greatest increase of the function
III - Negative gradients points in the direction of the greatest decrease of the function
IV - Hessian is a non-symmetric matrix

- ☐ I
☐ II and III
☒ I and IV
☐ III and IV

Yes, the answer is correct.



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Score: 1

Accepted Answers:

I and IV

7) The solution to an unconstrained optimization problem is always the same as the solution to the constrained one. **1 point**

☐ True

☒ False

Yes, the answer is correct.

Score: 1

Accepted Answers:

False

8) A manufacturer incurs a monthly fixed cost of \$7350 and a variable cost, **1 point**
 $C(m) = 0.001m^3 - 2m^2 + 324m$ dollars. The revenue generated by selling these units is,
 $R(m) = -6m^2 + 1065m$. How many units produced every month (m) will generate maximum profit?

☐

$m = 46$

☒

$m = 90$

☐

$m = 231$

☐

$m = 125$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$m = 90$

9) Consider an optimization problem $\min_{x_1, x_2} x^2 - xy + y^2$ subject to the constraints **1 point**

$$2x + y \leq 1$$

$$x + 2y \geq 2$$

$$x \geq -1$$

Find the lagrangian function for the above optimization problem.

☒

$$L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$$

☐

$$L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(x + 2y - 2) + \mu_3(-x - 1)$$

☐

$$L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(x + 2y - 2) + \mu_3(x + 1)$$

☐

$$L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(1 - 2x - y) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$$L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$$

